

Review class 1 - solutions

① $X_1, \dots, X_n \sim N(\theta, 4)$

a) 90% confidence interval for the mean with known variance:

$$\begin{aligned} & \left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} ; \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right] = \\ & = \left[1.45 - \frac{0.57}{4} t_{0.05}(15) ; 1.45 + \frac{0.57}{4} t_{0.05}(15) \right] = \\ & = \left[1.45 - \frac{0.57}{4} \cdot 1.753 ; 1.45 + \frac{0.57}{4} \cdot 1.753 \right] \approx \\ & \approx [1.45 - 0.22 ; 1.45 + 0.22] = \underline{[1.23 ; 1.67]} \end{aligned}$$

b) with probability 90% true mean belongs to the interval $[1.23; 1.67]$, but $2 \notin [1.23; 1.67]$

So, H_0 is wrong with probability at least 90%

$\Rightarrow$ we should reject H_0 .

c) For the known variance $\sigma^2 = 0.5$,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim Z \sim N(0,1)$$

$$Z = \frac{1.45 - \mu}{0.5/4} = 8(1.45 - \mu)$$

Confidence of $[5/4; 13/8]$ is

$$P(\mu \in [\frac{5}{4}; \frac{13}{8}]) = P(-8\mu \in [-13, -10]) \approx$$

$$P(8 \cdot (1.45 - \mu) \in [-1, -2]) = P(Z \in [-1, 2]) =$$

$$\begin{aligned}
 &= P(Z \leq 2) - P(Z \leq -1) \approx 0.9772 - (1 - P(Z \leq 1)) = \\
 &= 0.9772 - 1 + 0.8413 \approx 0.98 + 0.84 - 1 = \\
 &= \boxed{0.82}
 \end{aligned}$$

② $H_0: \mu_0 = 20; \sigma_0 = 4$

$H_1: \mu_1 = 25, \sigma_1 = 5$

$$\begin{aligned}
 0.05 = \alpha &= P(\bar{X} > 8 | H_0) = P\left(\frac{\bar{X} - \mu_0}{\frac{\sigma_0}{\sqrt{n}}} > \frac{8 - \mu_0}{\frac{\sigma_0}{\sqrt{n}}} | H_0\right) = \\
 &= P\left(Z > \frac{8 - 20}{\frac{4}{\sqrt{16}}}\right) \\
 &\quad Z \sim N(0,1)
 \end{aligned}$$

$$\frac{8 - 20}{\frac{4}{\sqrt{16}}} = Z_{0.05} = 1.64$$

$$8 = 8 \cdot 1.64 + 20 \approx \underline{93}$$

False acceptance probability = $\beta =$

$$= P(\bar{X} \leq 8 | H_1) = P\left(\frac{\bar{X} - \mu_1}{\frac{\sigma_1}{\sqrt{n}}} \leq \frac{8 - \mu_1}{\frac{\sigma_1}{\sqrt{n}}} | H_1\right)$$

$$\begin{aligned}
 &= P\left(Z \leq \frac{8 - 25}{\frac{5}{\sqrt{16}}}\right) \approx \Phi\left(\frac{93 - 25}{10}\right) = \Phi(6.8) \approx \\
 &\quad Z \sim N(0,1) \quad \underline{0.242}
 \end{aligned}$$

$$③ X_1, \dots, X_n \sim \text{Poi}(\mu)$$

$$H_0: \mu = 0.5 \text{ vs } H_1: \mu > 0.5$$

By Neyman-Pearson lemma, critical region C is ~~uniformly~~ best for testing $H_0: \mu = 0.5$ vs $H_1: \mu = \mu_1 > 0.5$, if

$$\frac{L(0.5)}{L(\mu)} \leq k \text{ for any } x \in C$$

$$\begin{aligned} \frac{L(0.5)}{L(\mu)} &= \frac{0.5^{\sum x_i} \cdot e^{-n/2}}{x_1! \dots x_n!} \cdot \frac{x_1! \dots x_n!}{\mu_1^{\sum x_i} e^{-\mu_1 n}} = \\ &= \left(\frac{1}{2\mu_1}\right)^{n\bar{x}} \cdot e^{\mu_1 n - n/2} \leq k \end{aligned}$$

$$n\bar{x} \cdot \ln\left(\frac{1}{2\mu_1}\right) + (\mu_1 - \frac{1}{2})n \leq \ln k$$

$$n\bar{x} \cdot \ln\left(\frac{1}{2\mu_1}\right) \leq \ln k - (\mu_1 - \frac{1}{2})n$$

$$\left[\mu_1 > \frac{1}{2} \Rightarrow \frac{1}{2\mu_1} < 1 \Rightarrow \ln \frac{1}{2\mu_1} < 0 \right]$$

$$n\bar{x} \geq \frac{\ln k - (\mu_1 - \frac{1}{2})n}{\ln\left(\frac{1}{2\mu_1}\right)}$$

$$(*) \quad C = \left\{ \bar{x} \geq \frac{\ln k - (\mu_1 - \frac{1}{2})n}{\ln\left(\frac{1}{2\mu_1}\right) \cdot n} \right\} \text{ is best critical region for } H_0 \text{ vs } H_1$$

So, $\bar{x}$ is enough to define best critical regions for H_0 vs $H_1: \mu = \mu_1$, and uniformly best crit. region for H_0 vs H_1 .

b) All uniformly best critical regions can be defined of the form $\tilde{C} = \{\bar{x} \geq c\}$.

Indeed, for $\forall c$ we can choose $k \geq 0$ such that and any μ_1

$$x \in \tilde{C} = \{\bar{x} \geq c\} \text{ if and only if } \frac{L(0.5; x)}{L(\mu_1; x)} \leq k$$

/this is the definition of the $\tilde{C}$ being uniformly best/

Indeed, $\frac{L(0.5; x)}{L(\mu_1; x)} \leq k$ when

$$\bar{x} \geq \frac{\ln k - (\mu_1 - 0.5)n}{\ln(\frac{1}{2\mu_1})n} \quad (\text{by a)})$$

or when

$\bar{x} \geq c$, if we choose

$$k = \exp\left(\ln\left(\frac{1}{2\mu_1}\right)nc + n(\mu_1 - \frac{1}{2})\right)$$

This k is positive since exponent is always ≥ 0 .

c) $\alpha = 0.068$

kind: $10\bar{X} \sim \text{Poi}(5)$ given H_0

$n = 10$

~~$10\bar{X} \sim \text{Poi}(10 \cdot \frac{1}{2})$~~

Goal is find c , such that

$P(\bar{X} \geq c | H_0) = 0.068$

$P(10\bar{X} \geq 10c | H_0) = P(Z \geq 10c) = 0.068$
 $\sim \text{Poi}(5)$
 or $(P(Z \leq 10c) = 0.932)$

From table III (Appendix)

it gives

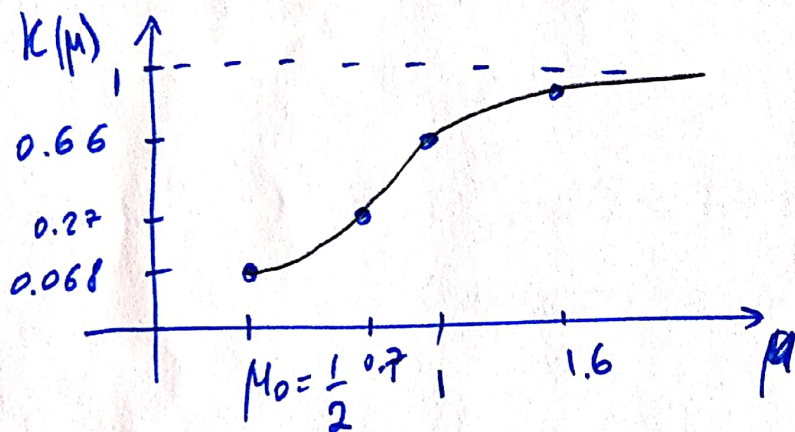
$10c = 8$ page 491, column 5.0

$$c = 0.8$$

Critical region $\{\bar{x} \geq 0.8\} \Rightarrow \text{reject } H_0$

(makes sense since $0.8 > 0.5 = \mu_0$)

d) Power function



$K(\mu)$ = probability to reject H_0 when parameter is μ

$$K(\mu) = P(\bar{x} \geq 0.8 \mid \mu) = P\left(\sum_{i=1}^2 Z_i \geq 8 \mid \mu\right)$$

e.g. $\mu = 1$

$$K(\mu) = P\left(\sum_{i=1}^2 Z_i \geq 8 \mid \mu\right) = 1 - 0.333 \sim 0.666$$

$\mu = 0.7$

$$K(\mu) = P\left(\sum_{i=1}^2 Z_i \geq 8 \mid \mu\right) = 1 - 0.729 \sim 0.27$$

$\mu = 1.6$

$$K(\mu) = P\left(\sum_{i=1}^2 Z_i \geq 8 \mid \mu\right) = 1 - 0.022 \sim 0.98$$

④ p-value is tail probability, so it can take values only in $[0, 1]$

We reject H_0 if and only if p-value $< \alpha$ = significance level = probability to reject H_0 if it is true

$P(\text{reject } H_0) \neq \alpha$, so just first sentence does not define α , so it is not enough to make a conclusion.

$$\begin{aligned} 0.5 = P(\text{reject } H_0) &= P(\text{reject } H_0 | H_0) \cdot P(H_0) + P(\text{reject } H_0 | H_1) \cdot P(H_1) \\ &= \alpha \cdot P(H_0) + (1 - \beta) \cdot P(H_1) = \\ &= \alpha \cdot (1 - 0.1) + (1 - \beta) \cdot 0.1 = \\ &= 0.9\alpha + 0.1(1 - \beta) \leq 0.9\alpha + 0.1 \quad \text{since } 0 \leq \beta < 1 \end{aligned}$$

$$0.5 \leq 0.9\alpha + 0.1$$

$$5 \leq 9\alpha + 1$$

$$4 \leq 9\alpha$$

$$\frac{4}{9} \leq \alpha$$

$$\alpha \geq \frac{4}{9} > \frac{3}{10} = 0.3 = \text{p-value, so}$$

H_0 should be rejected.